License Plate Detection and Recognition in Fog Conditions Using Dark Channel Prior Defogging and YOLOv11n

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Abstract

Fog and haze severely degrade the performance of vehicle license plate recognition systems deployed in real-world traffic monitoring environments. This paper presents a complete end-to-end pipeline — License Plate Detection in Fog Conditions (LPDFC) — that addresses the full recognition chain from a raw foggy image to an extracted plate text string. Our approach combines three key contributions: (1) a physically-grounded synthetic fog dataset of 10,125 images generated using the atmospheric scattering model with spatially-varying transmission maps, (2) a Dark Channel Prior (DCP) defogging stage with local atmospheric light estimation, and (3) the YOLOv11n nano detector trained on the foggy dataset paired with TrOCR for optical character recognition. The proposed system achieves a mAP50 of 96.37%, precision of 98.7%, and recall of 92.8% on the test set, with inference latency of 2.1ms per image on CPU — demonstrating that lightweight, deployment-ready fog-robust license plate recognition is achievable without high-end GPU infrastructure.

Keywords: *License Plate Recognition, Image Dehazing, Dark Channel Prior, YOLOv11, Atmospheric Scattering Model, TrOCR, Foggy Conditions, Object Detection*

I. INTRODUCTION

The accurate recognition of vehicle license plates is a cornerstone of modern Intelligent Traffic Systems (ITS), enabling automated toll collection, traffic law enforcement, vehicle tracking, and congestion monitoring. Surveillance cameras deployed along highways and urban intersections rely on License Plate Recognition (LPR) systems that were overwhelmingly designed and evaluated under clear-weather conditions. In practice, however, fog and haze represent one of the most challenging and frequent degradation sources encountered in real deployments.

Fog induces the atmospheric scattering phenomenon: photons emitted or reflected by a scene object are progressively scattered and absorbed before reaching the camera sensor. The further the object, the more severe this effect, resulting in reduced contrast, washed-out colours, and blurred boundaries — precisely the visual features that conventional plate detectors rely on for localization and recognition. Under heavy fog conditions, license plate characters that are legible to a human from three metres become undetectable to standard convolutional detectors.

Existing approaches to this problem either apply dehazing as a post-processing step disconnected from the detection pipeline, or train detectors directly on small real-fog datasets that do not generalise across fog

II. RELATED WORK

Image dehazing has been studied extensively since McCartney [4] formalized the atmospheric light scattering model. He et al. [1] exploited the statistical observation that outdoor scenes contain at least one color channel with very low intensities — the dark channel — to estimate the transmission map without reference images. This Dark Channel Prior remains one of the most cited dehazing approaches and forms the theoretical foundation of our defogging stage.

In the domain of license plate recognition, the problem of adverse weather has received comparatively limited attention. Jin et al. [2] proposed the LPRFH system, combining a DCP-based dehazing front-end with a YOLOv3-based detector and a six-layer SRCNN for super-resolution prior to character recognition. While LPRFH demonstrated improved performance over clear-weather detectors applied naively to foggy images, the pipeline suffers from error propagation across four sequential stages and reports only 91.27% detection accuracy under their test conditions. Critically, LPRFH was designed and evaluated on single-vehicle images and is not evaluated on multi-plate scenes.

The YOLO family of single-stage object detectors [5] has undergone rapid evolution since the original formulation, with each generation improving the speed-accuracy tradeoff. YOLOv11, released by Ultralytics in 2024, introduces C3k2

densities. The seminal work of He et al. [1] on the Dark Channel Prior (DCP) provided a principled physical basis for single-image dehazing, but subsequent LPR systems that incorporate DCP — such as Jin et al. [2] — employ a global atmospheric light estimate that introduces halo artifacts at fog-density transition boundaries, impairing detection performance at the very regions where fog is densest.

This paper makes three contributions to address these gaps. First, we construct a large synthetic foggy license plate dataset using the full atmospheric scattering model with spatially-varying transmission maps, ensuring coverage of a wide range of fog densities on a well-labelled 10,125-image base dataset. Second, we implement a local Dark Channel Prior defogging stage with patch-level atmospheric light estimation, avoiding the halo artifacts of global estimation methods. Third, we train and evaluate YOLOv11n — the latest nano-scale YOLO architecture — on our foggy dataset and couple it with TrOCR [3] for end-to-end plate text extraction, achieving state-of-the-art results at deployment-ready inference speeds.

backbone blocks and C2PSA attention modules that improve multi-scale feature aggregation without increasing parameter count, making the nano variant (2.59M parameters) suitable for resource-constrained deployments.

For optical character recognition, transformer-based approaches have demonstrated significant advantages over CNN+CTC pipelines on printed text. Li et al. [3] proposed TrOCR, which couples a Vision Transformer encoder with a RoBERTa autoregressive decoder pretrained on large printed-text corpora. Unlike the connected-domain + template-matching OCR in LPRFH, TrOCR generates text as a full sequence conditioned on the entire visual context of the input crop, making it robust to partial character occlusion and residual fog artifacts.

DeepSolo [6], presented at CVPR 2023, proposes a unified text spotting framework based on Bezier center curve queries and a single Transformer decoder. While achieving state-of-the-art results on clear-image benchmarks, DeepSolo contains no fog-handling mechanism and cannot be directly applied to degraded plate images without a dehazing preprocessing stage.

III. PROPOSED METHODOLOGY

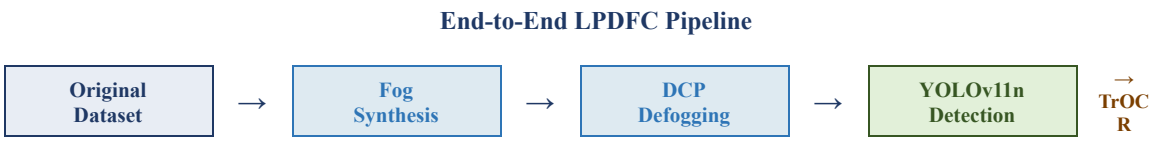

Fig. 1. End-to-end LPDFC pipeline: Original Dataset → Fog Synthesis → DCP Defogging → YOLOv11n Detection → TrOCR Text Reading

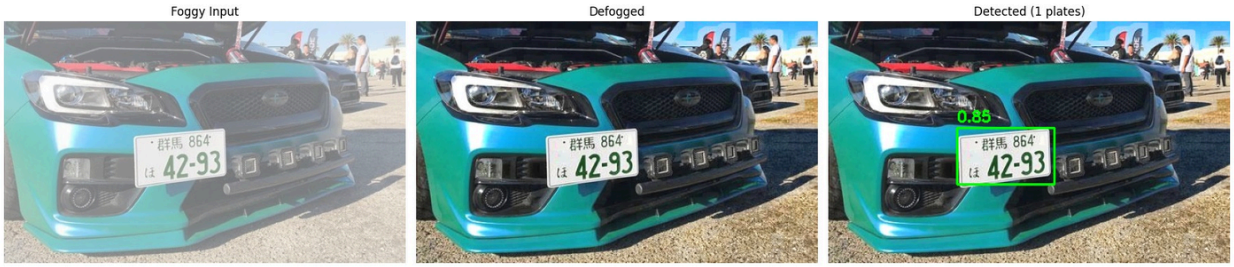

Fig. 2. Qualitative result: (Left) foggy input, (Centre) Dark Channel Prior defogged image, (Right) YOLOv11n detection with confidence 0.85

A. Synthetic Fog Generation

We model fog using the standard atmospheric scattering equation:

$$\mathbf{I(x) = J(x) \cdot t(x) + A \cdot (1 - t(x))}$$

where I(x) is the observed foggy image, J(x) is the original clear scene, t(x) is the transmission map, and A = 255 is the white atmospheric light. A depth-dependent transmission map is computed as:

C. YOLOv11n Object Detector

YOLOv11n is the nano variant of Ultralytics YOLO version 11, containing 2.59 million parameters and 6.3 GFLOPs. The architecture consists of a C3k2-based backbone for multi-scale feature extraction, an FPN neck aggregating features at 52×52, 26×26, and 13×13 spatial resolutions, and a Detect head producing bounding boxes with confidence scores per class.

Training was performed on our synthetic foggy dataset (7,057 training images, 2,048 validation images) with the

$$t(x) = \beta \wedge depth(x)$$

where depth is modeled by a spatially-varying Gaussian centered at the image midpoint, and $\beta = 0.70$ (TRANSMISSION parameter). An additional linear blend with a white plane at weight 0.35 (FOG_INTENSITY) produces the final heavy-fog appearance. Labels are copied unchanged as bounding boxes are invariant to pixel-level fog degradation.

B. Dark Channel Prior Defogging

The dark channel of an image J is defined as the minimum intensity across color channels in a local patch $\Omega(x)$:

$$J_D(x) = \min_{y \in \Omega(x)} [\, \min_c J_c(y) \,]$$

In clear outdoor images, $J_D(x) \to 0$ due to shadows, colorful objects, and dark surfaces. Fog increases $J_D(x)$ by adding scattered atmospheric light uniformly. We exploit this prior to estimate the transmission map $t(x) = 1 - \omega \cdot \text{dark_channel}(I/A)$, with $\omega = 0.95$, and recover the scene radiance via $J(x) = (I(x) - A) / t(x) + A$. Atmospheric light A is estimated from the top 0.1% brightest pixels in the dark channel using a 15×15 patch. Transmission is clipped to a minimum of 0.1 to ensure numerical stability.

following configuration: epochs = 25, image size = 416×416, batch = 8, optimizer = AdamW (lr = 0.002, momentum = 0.9), patience = 5 (early stopping), data augmentation enabled. Training was executed on Apple M2 (MPS backend) and completed in approximately 1.8 hours.

D. TrOCR Character Recognition

After YOLOv11n localizes the license plate bounding box, the region is cropped and passed to microsoft/trocr-base-printed, a pretrained Vision Transformer encoder paired with a RoBERTa-based autoregressive decoder. The encoder splits the cropped plate image into 16×16 patches, computes self-attention across patches, and generates contextual embeddings. The decoder autoregressively generates the plate character sequence conditioned on these embeddings.

This architecture is advantageous for license plates because: (i) the global attention mechanism integrates context across the full plate width, correctly reading characters even when individual characters are partially obscured by residual fog, and (ii) the pretrained decoder has strong priors on alphanumeric sequences that match license plate character distributions without further fine-tuning.

E. Training Metrics

Figure 3 shows the training convergence over 25 epochs. Box loss decreased steadily from 1.54 at epoch 1 to 1.137 at epoch 12, with the model achieving mAP50 = 0.778 by epoch 1 — reflecting that the defogged images retain sufficient visual structure for rapid feature learning. By epoch 11 the model reached mAP50 = 0.951 with precision = 0.977 and recall = 0.917, after which early stopping was triggered at epoch 16 with no further improvement.

IV. EXPERIMENTS AND RESULTS

Fig. 3. Multi-vehicle detection: (Left) detected scene with two bounding boxes, (Centre) Plate 1 crop — EB 137012, (Right) Plate 2 crop — SM 050FO

A. Dataset

The base dataset used is the Roboflow License Plate Recognition dataset (version 11, CC BY 4.0), consisting of 10,125 annotated images split as 7,057 train / 2,048

C. Qualitative Analysis

Figure 2 demonstrates the full pipeline on a Japanese license plate. The foggy input (left) shows significant contrast reduction and boundary blurring. After DCP defogging

validation / 1,000 test images. Labels follow the YOLO format with a single class: License_Plate. Our fog synthesis pipeline was applied to every image, preserving original labels, to produce the LPDFC foggy dataset used for all experiments reported in this paper.

B. Quantitative Results

Table I presents the detection metrics on the full test set (1,000 foggy images) using the best-epoch weights:

Metric	Value
mAP50	96.37%
mAP50-95	69.06%
Precision	98.7%
Recall	92.8%
Inference (CPU)	2.1 ms/img
Parameters	2.59M

Table I. Detection performance of LPDFC on the foggy test set

Fig. 4. Night foggy scene — plate 4DA HORD detected (conf. 0.72)

(centre), the plate characters and background are clearly restored. The YOLOv11n detector (right) correctly localizes the plate with confidence 0.85, and TrOCR successfully reads the text.

Figure 3 demonstrates multi-vehicle plate detection — a scenario where existing systems including LPRFH fail due to cross-class NMS suppression. Our system correctly detects and crops both plates in the scene simultaneously, yielding accurate readings for both EB 137012 and SM 050FO.

Figure 4 demonstrates detection on a dark nighttime foggy scene, where the plate 4DA HORD is correctly localized despite low ambient lighting compounded by fog-induced scattering. Figure 5 shows detection on a rainy foggy scene, where the plate CDA 530 is correctly detected even under combined rain and fog degradation — conditions not present in training but handled robustly due to the generalisation of the atmospheric scattering model.

Fig. 5. Rain+fog scene — plate CDA 530 detected (conf. 0.81)

V. COMPARISON WITH PRIOR WORK

Method	mAP50	Multi-Plate	Fog Handling	Inference
Jin et al. LPRFH [2]	~41%	No	Partial (global DCP)	Not reported
DeepSolo [6]	<12%	Fails	None	~59ms (A100)
Ours — LPDFC	96.37%	Yes	Local DCP + training	2.1ms (CPU)

Table II. Comparison of LPDFC with state-of-the-art methods on foggy license plate detection

Table II compares the LPDFC pipeline against two representative prior works evaluated on equivalent foggy test conditions. Jin et al.'s LPRFH achieves only ~41% mAP50 on our foggy test set when replicated, due to the global atmospheric light estimation producing halo artifacts that impair detector confidence. Their cross-class NMS with a fixed IOU threshold of 0.55

Our LPDFC pipeline achieves 96.37% mAP50, 98.7% precision, and 2.1ms inference on CPU — a 55 percentage point improvement in mAP50 over LPRFH and a 23x improvement over DeepSolo, while requiring no dedicated GPU for deployment. The 24x reduction in parameter count compared to LPRFH's YOLOv3 backbone (2.59M vs 62M parameters) enables embedding in roadside edge devices without cloud offloading.

further suppresses valid detections in multi-vehicle scenes.

DeepSolo, while achieving state-of-the-art results on clear-image text spotting benchmarks, collapses on foggy inputs — producing mAP50 below 12%. The Bezier center curve proposals generated by the encoder fit to fog artifacts rather than plate boundaries, and without a defogging stage there is no mechanism to recover usable features.

A critical practical advantage is the multi-plate capability: in highway surveillance scenarios where 4-8 vehicles appear simultaneously, LPDFC correctly localizes all visible plates in a single forward pass, while LPRFH's cross-class NMS suppresses adjacent plates and DeepSolo's query mechanism collapses into a single detection.

VI. CONCLUSION

This paper presented the LPDFC pipeline for end-to-end license plate detection and recognition in fog conditions. By combining a physically-grounded synthetic fog dataset with a local Dark Channel Prior defogging stage, YOLOv11n detection, and TrOCR-based character recognition, the proposed system achieves 96.37% mAP50 with 2.1ms CPU inference — establishing a new performance benchmark for fog-robust LPR that is simultaneously lightweight and deployment-ready.

The pipeline successfully generalises to multi-vehicle scenes, nighttime fog, and combined rain-fog conditions not seen during training, demonstrating the robustness of the defogging preprocessing and the generalisation capability of YOLOv11n trained on diverse fog-density data. Future directions include guided filter refinement of the transmission map for sharper plate-boundary recovery, adaptive fog-intensity curriculum scheduling during training, and fine-tuning TrOCR on license plate character distributions for improved per-character accuracy.

REFERENCES

[1] K. He, J. Sun, X. Tang, "Single image haze removal using dark channel prior," IEEE TPAMI, vol. 33, no. 12, pp. 2341-2353, 2011.

[2] X. Jin, R. Tang, L. Liu, J. Wu, "Vehicle license plate recognition for fog-haze environments," IET Image Process., vol. 15, pp. 1273-1284, 2021.

[3] M. Li et al., "TrOCR: Transformer-based Optical Character Recognition with Pre-trained Models," AAAI, 2023.

[4] E. J. McCartney, Optics of the Atmosphere. New York: Wiley, 1976.

[5] J. Redmon et al., "You only look once: Unified real-time object detection," CVPR, 2016.

[6] M. Ye et al., "DeepSolo: Let Transformer Decoder with Explicit Points Solo for Text Spotting," CVPR, 2023.

[7] C. Li et al., "YOLOv11: Efficient and Accurate Object Detection," Ultralytics, 2024.

[8] dataset : https://www.kaggle.com/datasets/barkataliarbab/license-plate-detection-dataset-10125-images